



Building Java Programs
A BACK TO BASICS APPROACH 5e

Computer Science

Implementation - Heap

Instructor: Varik Hoang

Platform: HuskyMap

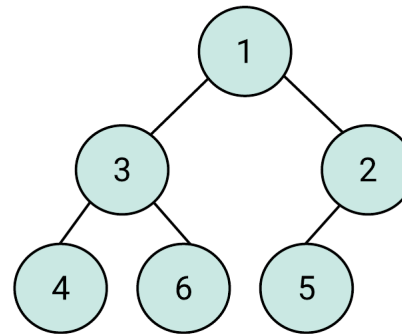


Stuart Reges / Marty Stepp

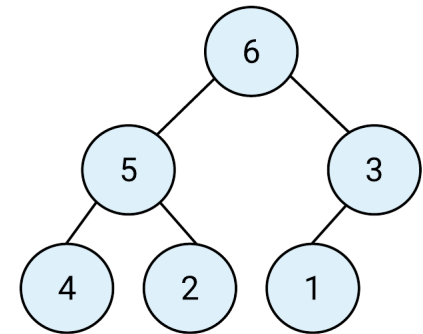


Objectives

- Use an array to represent a heap
- Add an entry to an array-based heap
- Remove the root of an array-based heap
- Create a heap from given entries



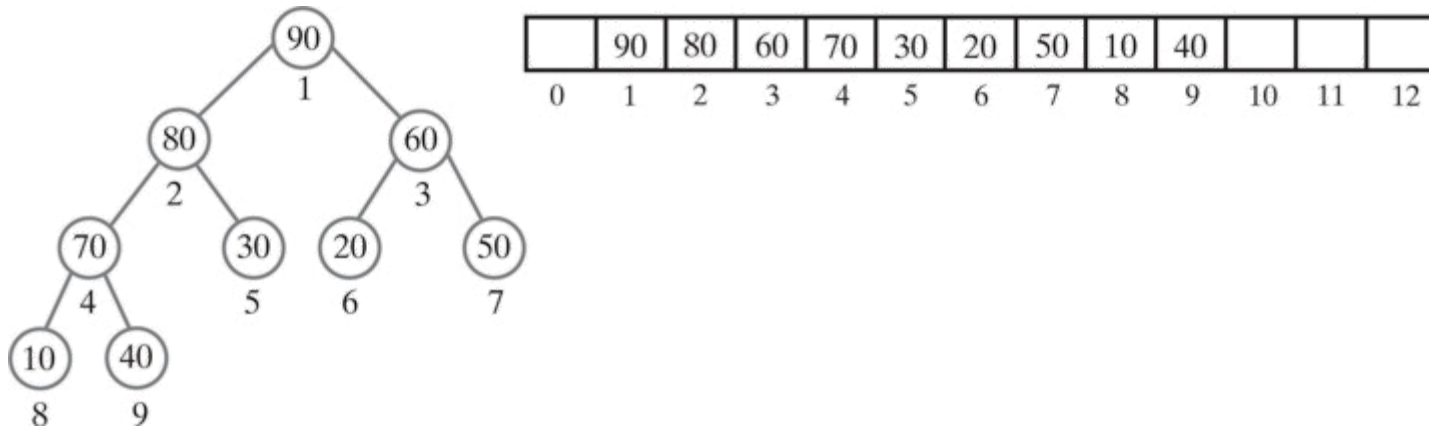
Min heap



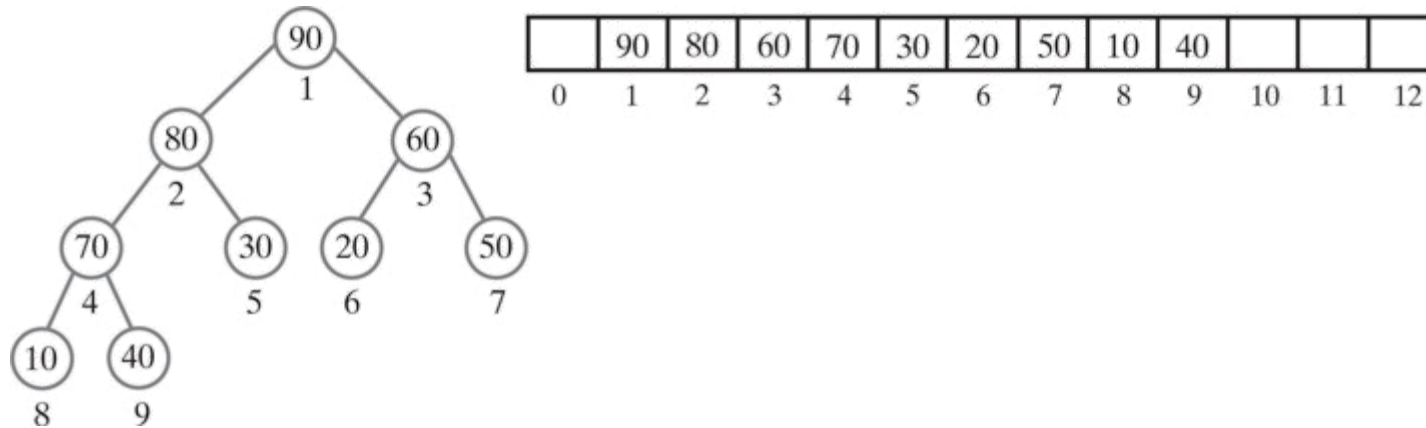
Max Heap

Using an Array to Represent a Heap

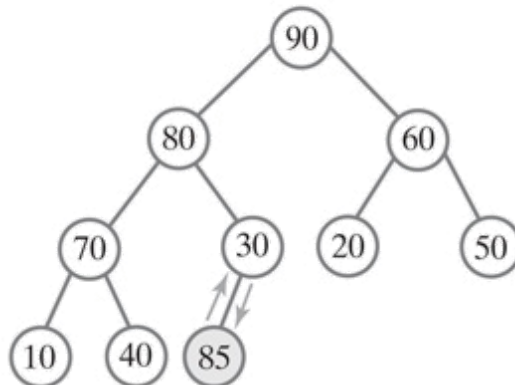
- A complete tree is full to its next-to-last level
 - Its leaves on the last level are filled in from left to right
- Suppose that we place item in tree into consecutive array locations
- Locate either the children/parent of any node with simple computation
- Enable us to quickly locate the data in a node's parent or children



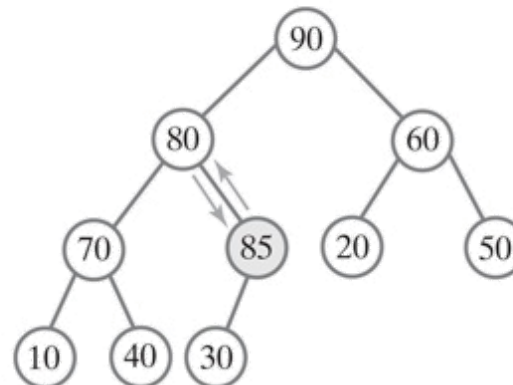
MyMaxHeap - Adding an Entry



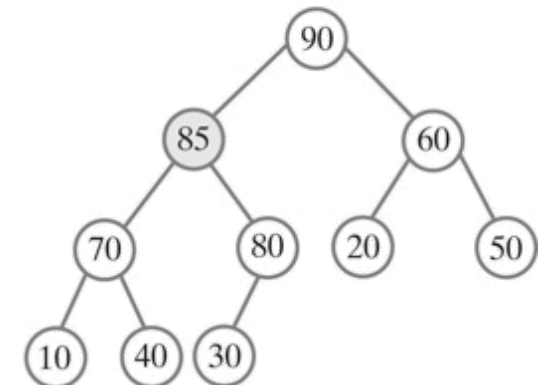
(a) Add 85 as the next leaf.
Then swap it with its parent, 30



(b) Swap 85 with its parent, 80

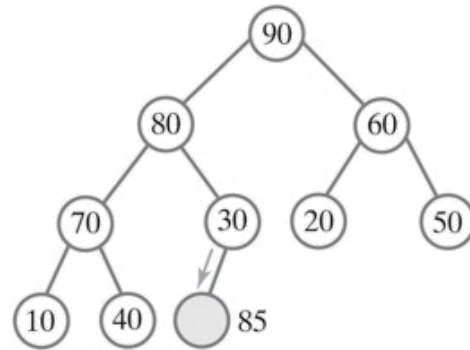


(c) The result is a max heap

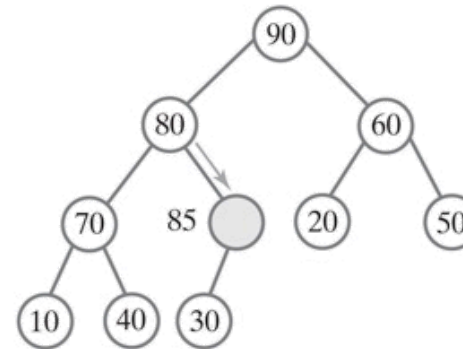


MyMaxHeap - Adding an Entry (cont.)

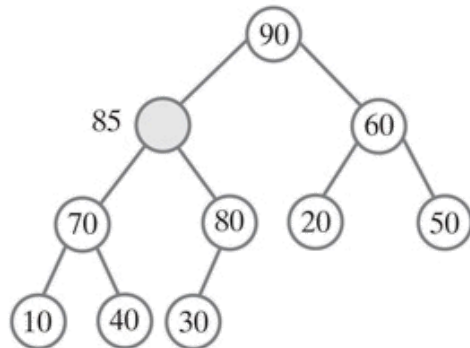
- (a) Identify the location of a new leaf.
85 is larger than 30, which is in this leaf's parent, so move 30 to the new leaf



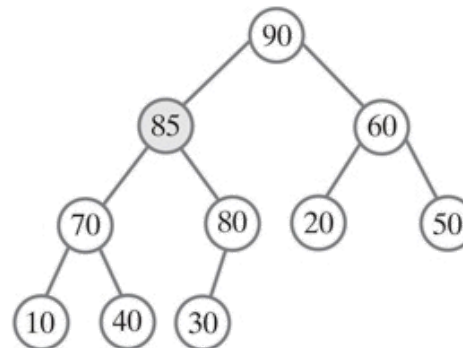
- (b) 85 is larger than 80, which is in the empty node's parent, so move 80 to the empty node



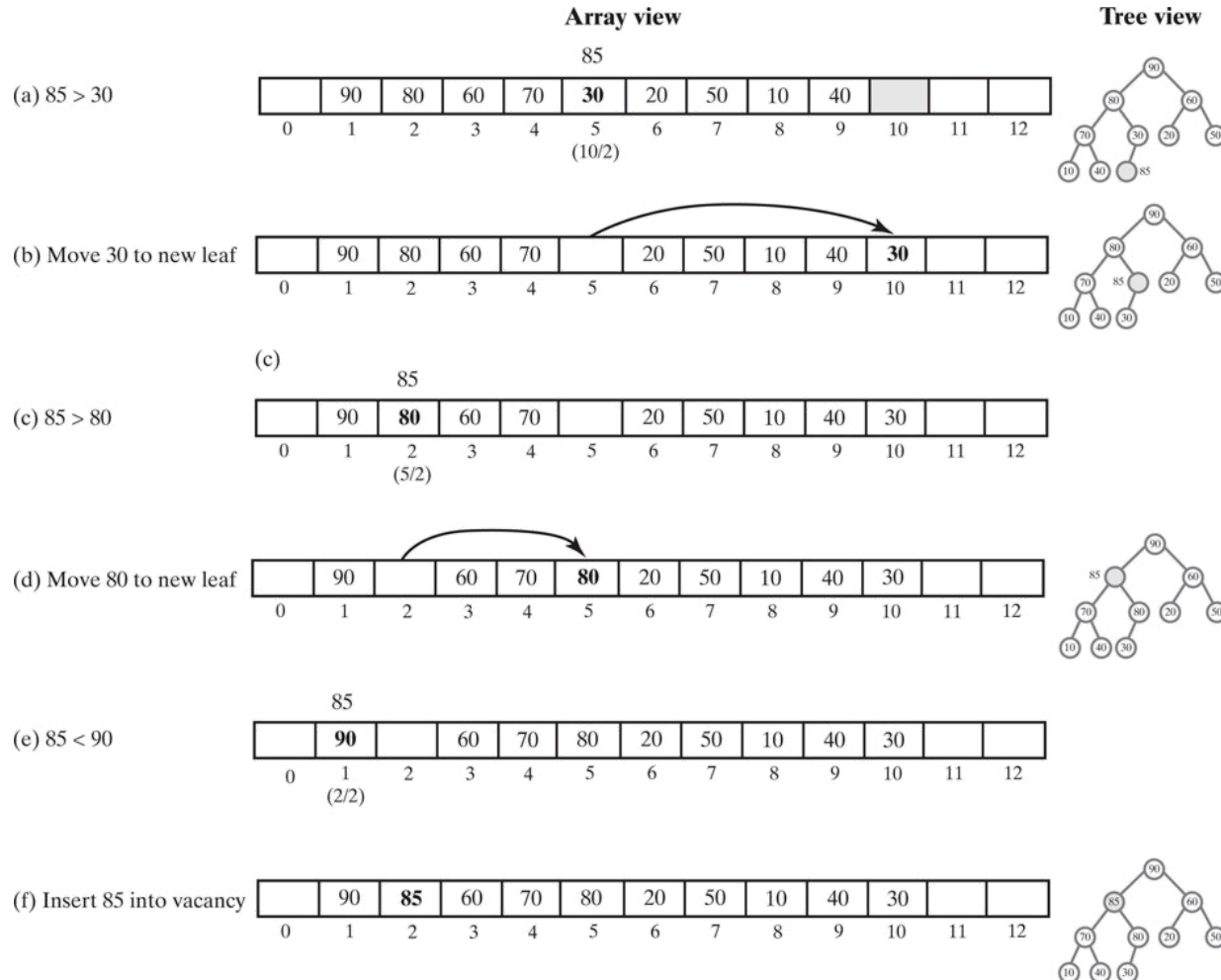
- (c) 85 is less than 90, which is in the empty node's parent, so place 85 into the empty node



- (d) The result is a maxheap

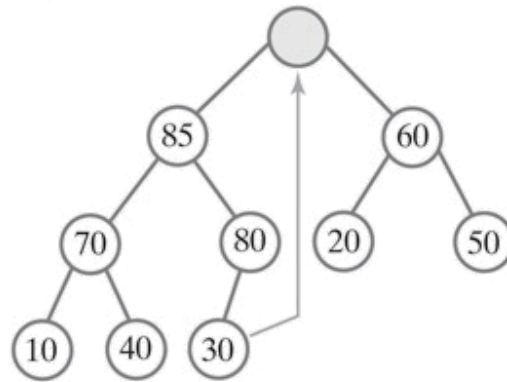


MyMaxHeap - Adding an Entry (cont.)

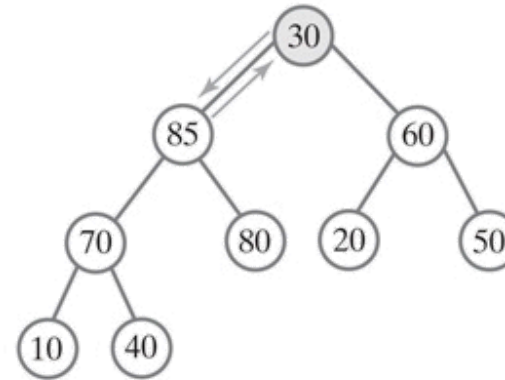


MyMaxHeap - Removing the Root

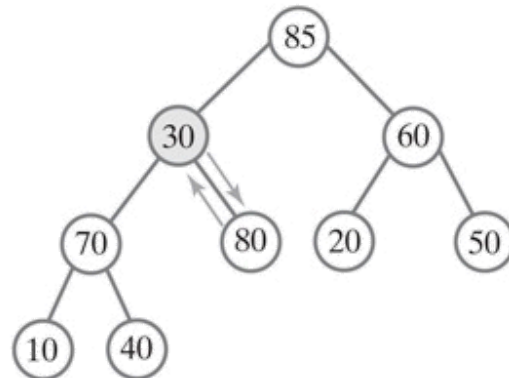
(a) Replace the root's entry with the last leaf's data



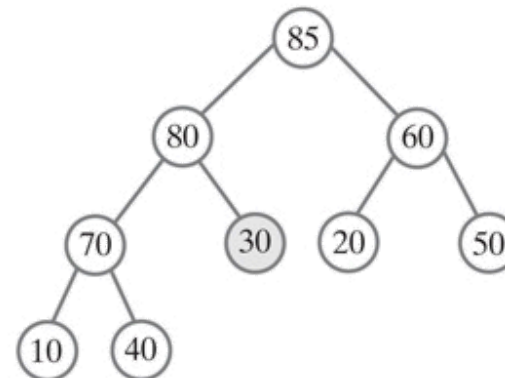
(b) Delete the last leaf; swap 30 with its largest child, 85



(c) Swap 30 with its largest child, 80

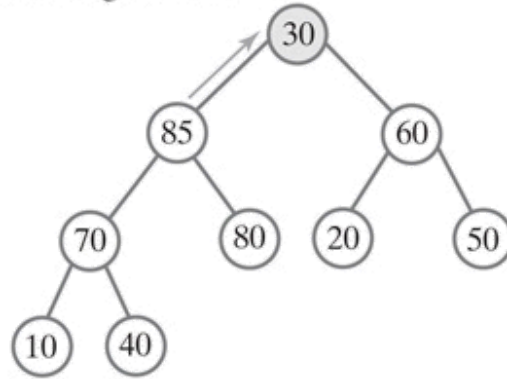


(d) The result is a maxheap

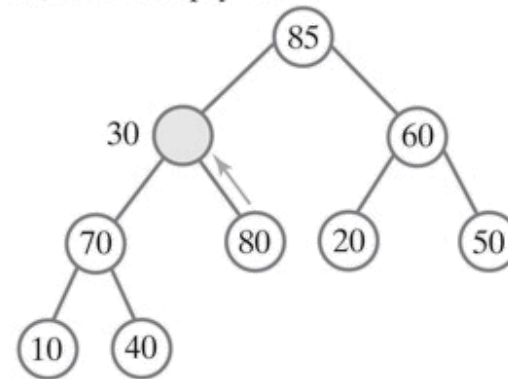


MyMaxHeap - Removing the Root (cont.)

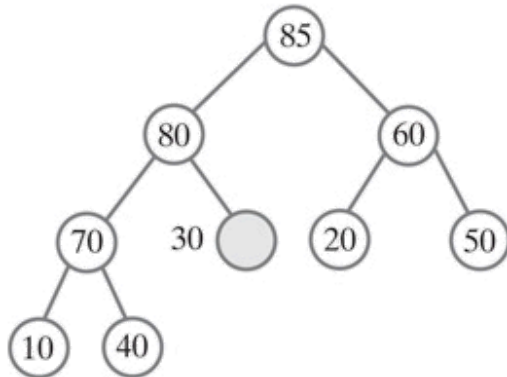
(a) Copy 30 and replace it with the root's largest child



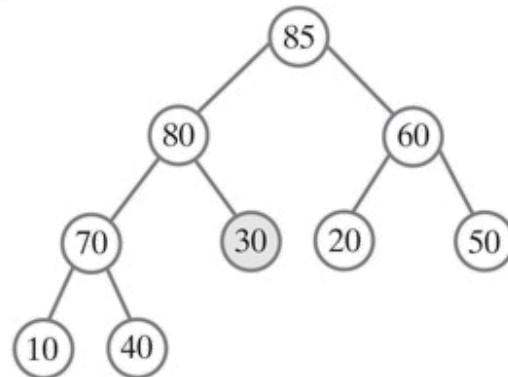
(b) Move the empty node's larger child, 80, to the empty node



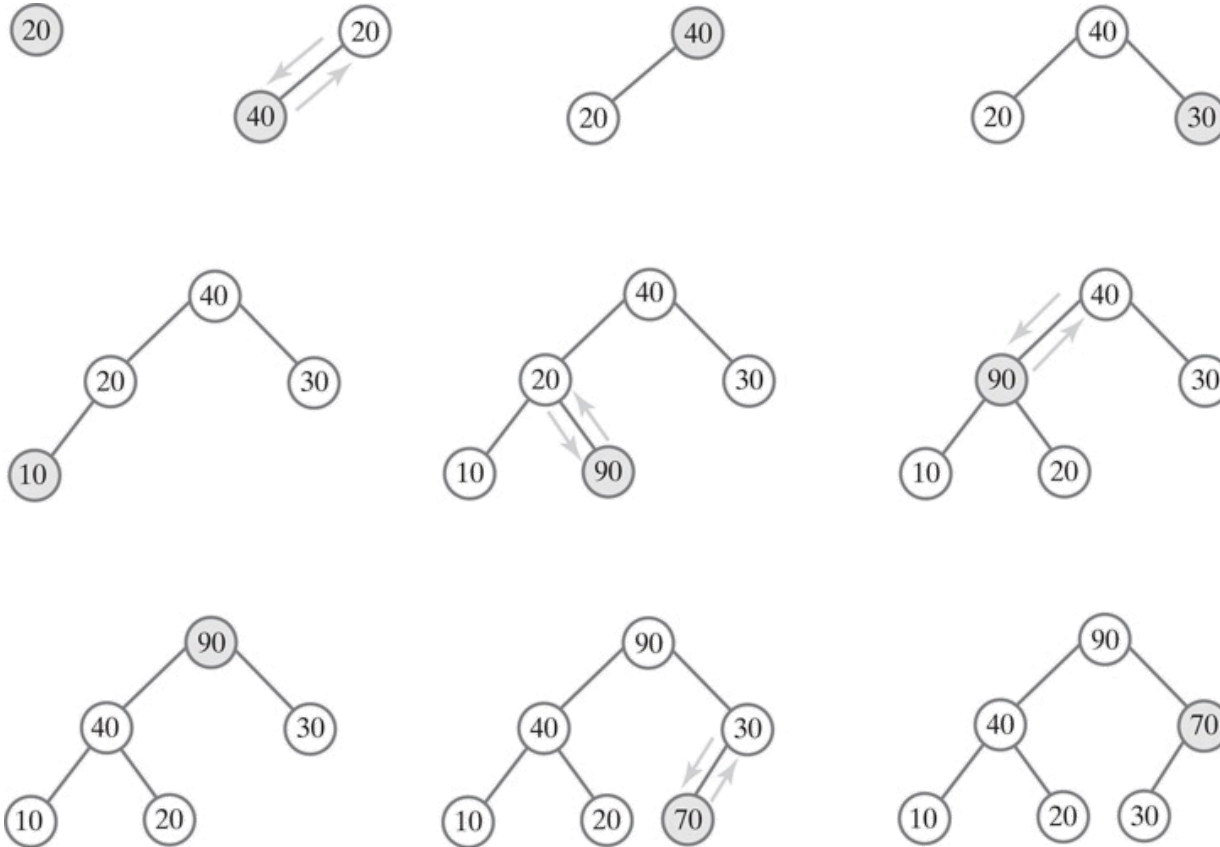
(c) Place 30 into the vacant leaf



(d) The result is a maxheap

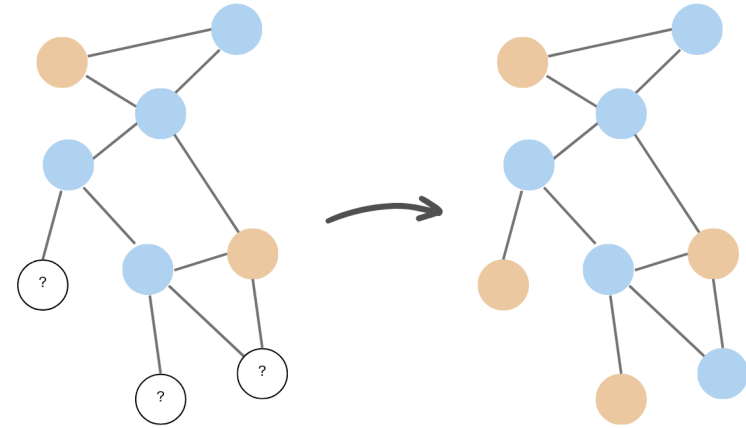


Creating a Heap



Applications

- Priority Queues
- Heap Sort
- Memory Allocation
- Graph Algorithms
- Job Scheduling
- Event-Driven Simulations
- Data Compression Algorithms



Questions & Answer





Thank You!

A black fountain pen with silver-colored accents and a silver nib is positioned diagonally below the end of the 'Thank You!' text. The pen is lying horizontally, pointing towards the bottom right, and has a soft shadow beneath it.